

Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

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Chapter 1

Low Dimensional Asymptotic Normality

1.1 Basic Setup

Given a "high dimensional" data set $\{\mathbf{x}_i\}_{i=1}^n$ where $\mathbf{x}_i \in \mathbb{R}^p$ for all $i = 1, \dots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean $\mathbf{0}$), the probabilistic principal component analysis model is a statistical model that treats $\mathbf{x}_i$'s as realizations of independent random variables $\mathbf{X}_i = \mathbf{W}\mathbf{Z}_i + \boldsymbol{\epsilon}_i$, where $\mathbf{W}$ is an unknown $p \times q$ loading matrix, $\mathbf{Z}_i \sim N_q(\mathbf{0}, \mathbf{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon}_i \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and $p(\mathbf{x}; \theta)$ for the density of $\mathbf{X}_1$, namely the density of $N_p(\mathbf{0}, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 \mathbf{I}_p)$.

We now setup the quotient space structure. Write Θ for the collection of all tuples $(\mathbf{W}, \sigma^2)$ such that $\mathbf{W}$ is a real $p \times q$ matrix with $\text{rank}(\mathbf{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$, hence a Riemannian manifold with the standard Euclidean metric

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = \text{tr}(\mathbf{V}\mathbf{U}^T) + cb, \quad (1.1)$$

whenever $(\mathbf{A}, s) \in \Theta$, $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$. Next, we define $(\mathbf{A}, s)$ and $(\mathbf{B}, t) \in \Theta$ to be equivalent when they give the same density, namely when $\mathbf{A}\mathbf{A}^T + s\mathbf{I}_p = \mathbf{B}\mathbf{B}^T + t\mathbf{I}_p$, and write $(\mathbf{A}, s) \sim (\mathbf{B}, t)$. The equivalence classes are denoted by $[(\mathbf{A}, s)]$ or more briefly $[\mathbf{A}, s]$. Let us denote by $O(q)$ the collection of all $q \times q$ real orthogonal matrices. It is shown in Datta's thesis that $[\mathbf{A}, s] = \{(\mathbf{A}\mathbf{Q}, s) \mid \mathbf{Q} \in O(q)\}$. Unless otherwise mentioned, $(\mathbf{A}, s) \in \Theta$ is the default representative we pick when we write an equivalence class as $[\mathbf{A}, s]$. Therefore, $M := \Theta / \sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee(2013)], Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension $pq + 1 - \frac{q(q-1)}{2}$ and $\pi : \Theta \rightarrow M$ with $\pi((\mathbf{A}, s)) = [\mathbf{A}, s]$ is a smooth submersion. Moreover, this action is verticle, transitive on fibers and is isomrtic with respect to (1.1). Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see [Lee(2018)], Theorem 2.28), meaning that

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = g_{[\mathbf{A}, s]}(d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c)), d\pi_{(\mathbf{A}, s)}((\mathbf{U}, b))), \quad (1.2)$$

whenever $(\mathbf{A}, s) \in \Theta$, $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$.

1.2 Exponential Map and Logarithmic Map Computation

In this section, we carry out some basic computations on (M, g) . Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and verticle tangent spaces. We begin with a few technical details in the computation.

Proposition 1. Fix $(\mathbf{A}, s) \in \Theta$. The verticle tangent space $V_{(\mathbf{A}, s)} := \ker(d\pi_{(\mathbf{A}, s)})$ equals to the collection of all tangent vectors of the form $(\mathbf{A}\Omega, 0) \in T_{(\mathbf{A}, s)}\Theta$, with Ω ranged over all $q \times q$ skew symmetric matrices. The horizontal tangent space $H_{(\mathbf{A}, s)} := V_{(\mathbf{A}, s)}^\perp$ equals to the collection of all tangent vectors of the form $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $q \times q$ matrices $\mathbf{V}$ satisfying $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$.

Proof. We start with computing the verticle tangent space. Define function $\Phi : \Theta \rightarrow \mathbb{R}^{p \times p}$ by setting, for every $(\mathbf{A}, s) \in \Theta$:

$$\Phi((\mathbf{A}, s)) = \mathbf{A}\mathbf{A}^T + s\mathbf{I}_p. \quad (1.3)$$

We claim that

$$\ker(d\pi_{(\mathbf{A}, s)}) = \ker(d\Phi_{(\mathbf{A}, s)}).$$

Before proving the claim, let us first study $\ker(d\Phi_{(\mathbf{A}, s)})$. For $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ by setting $\gamma(t) = (\mathbf{A} + t\mathbf{V}, s + tc)$. Then:

$$d\Phi_{(\mathbf{A}, s)}((\mathbf{V}, c)) = \frac{d}{dt} \big|_{t=0} \Phi(\gamma(t)) = \mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + 2sc\mathbf{I}_p.$$

If $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A}, s)})$, then

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + 2sc\mathbf{I}_p = \mathbf{0}. \quad (1.4)$$

If $\mathbf{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\mathbf{A}$, then $\mathbf{A}^T \mathbf{u} = \mathbf{0}$ and

$$0 = \mathbf{u}^T (\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + 2sc\mathbf{I}_p) \mathbf{u} = 2sc\mathbf{u}^T \mathbf{u}.$$

Given that $s > 0$, we must have $c = 0$ and (1.4) reduces to

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T = \mathbf{0}. \quad (1.5)$$

Apply $\mathbf{u}$ to the right to get:

$$\mathbf{A}\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Given that $\mathbf{A}$ has full rank, rank nullity theorem forces

$$\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Therefore, we have proven that every vector $\mathbf{u}$ orthgonal to columns vectors of $\mathbf{A}$ will also be orthgonal to column vectors of $\mathbf{V}$. This means $\mathbf{V} = \mathbf{A}\Omega$ for some $q \times q$ matrix Ω . Substitute this back to (1.5) gives

$$\mathbf{A}(\Omega^T + \Omega)\mathbf{A}^T = \mathbf{0}.$$

Applying $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ to the left and $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$ to the right yields $\Omega^T + \Omega = \mathbf{0}$. Therefore, Ω is skew-symmetric. Conversely, (1.5) is satisfied by $\mathbf{V} = \mathbf{A}\Omega$ with any skew-symmetric matrix Ω . We have proven that $\ker(d\Phi_{(\mathbf{A}, s)})$ equals to the collection of all tangent vectors of the form $(\mathbf{A}\Omega, 0) \in T_{(\mathbf{A}, s)}\Theta$, with Ω ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove the claim. Let us define function $F : M \rightarrow \mathbb{R}^{p \times p}$ by setting $F([\mathbf{A}, s]) = \Phi((\mathbf{A}, s))$. Clearly, F satisfies $F \circ \pi = \Phi$ and is hence a smooth injective function (here the smoothness of F is implied by for example [Lee(2013)] Theorem 4.29). By the chain rule, we have

$$d\Phi_{(\mathbf{A}, s)} = dF_{[\mathbf{A}, s]} \circ d\pi_{(\mathbf{A}, s)}. \quad (1.6)$$

It follows immediately that $\ker(d\Phi_{(\mathbf{A},s)}) \subseteq \ker(d\pi_{(\mathbf{A},s)})$. The other direction will follow if we can prove that $dF_{[\mathbf{A},s]}$ is injective. To do so, suppose that $v \in T_{[\mathbf{A},s]}M$ is such that $dF_{[\mathbf{A},s]}(v) = 0$. Let $(\mathbf{V}, c) \in T_{(\mathbf{A},s)}\Theta$ be the (unique) horizontal lift of v at $(\mathbf{A}, s)$, meaning that $(\mathbf{V}, c)$ is the unique horizontal tangent vector at $(\mathbf{A}, s)$ such that $v = d\pi_{(\mathbf{A},s)}((\mathbf{V}, c))$. (1.6) then implies that $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A},s)})$ and the proceeding discussion shows that $c = 0$ and $\mathbf{V} = \mathbf{A}\Omega$ for some skew-symmetric Ω . To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ defined by $\gamma(t) = (\mathbf{A}e^{t\Omega}, s)$. Clearly $\gamma(0) = (\mathbf{A}, s)$ and $\gamma'(0) = (\mathbf{A}\Omega, 0)$. Moreover, the matrix $e^{t\Omega}$ is always orthogonal for any t and any skew-symmetric Ω so that $\pi(\gamma(t)) = [\mathbf{A}, s]$ for all t . We have

$$v = d\pi_{(\mathbf{A},s)}((\mathbf{A}\Omega, 0)) = d\pi_{(\mathbf{A},s)}(\gamma'(0)) = \frac{d}{dt} \big|_{t=0} \pi(\gamma(t)) = \mathbf{0} \in T_{[\mathbf{A},s]}M.$$

Therefore, $dF_{[\mathbf{A},s]}$ is injective and the claim is proved. The verticle tangent space is therefore the same as $\ker(d\Phi_{(\mathbf{A},s)})$.

To compute the horizontal tangent space, any $(\mathbf{V}, c) \in V_{(\mathbf{A},s)}^\perp$ will satisfy

$$0 = g_{(\mathbf{A},s)}((\mathbf{V}, c), (\mathbf{A}\Omega, 0)) = \text{tr}(\mathbf{A}^T \mathbf{V} \Omega) \quad (1.7)$$

for all skew-symmetric Ω . Decomposing $\mathbf{A}^T \mathbf{V} = \mathbf{S} + \mathbf{K}$, where $\mathbf{S}$ is the symmetric part and $\mathbf{K}$ is the skew-symmetric part, we find out that

$$0 = \text{tr}(\mathbf{S}\Omega) + \text{tr}(\mathbf{K}\Omega) = 0 + \text{tr}(\mathbf{K}\Omega) = \text{tr}(\mathbf{K}\Omega).$$

Setting $\Omega = \mathbf{K}^T$ will yield that $\mathbf{K} = \mathbf{0}$. Therefore, $\mathbf{A}^T \mathbf{V}$ is a symmetric matrix. Conversely, any $\mathbf{V}$ such that $\mathbf{A}^T \mathbf{V}$ is symmetric will satisfy (1.7). The proof is complete. $\square$

Proposition 2. Given $[\mathbf{A}, s] \in M$ and $v \in T_{[\mathbf{A},s]}M$, geodesics in M starting at $[\mathbf{A}, s]$ with initial velocity v takes the format of curves $\gamma : I \rightarrow M$, where I is an open interval containing 0, defined by $\gamma(t) = [\mathbf{A} + t\mathbf{V}, s + tc]$, where $(\mathbf{V}, c) \in H_{(\mathbf{A},s)}$ is the horizontal lift of v at $(\mathbf{A}, s)$. Consequently, the exponential map is computed as $\text{Exp}_{[\mathbf{A},s]}(v) = [\mathbf{A} + \mathbf{V}, s + c]$, well-defined over all v with small norm.

Proof. First of all, let us quickly check that the curve γ is well defined. Indeed, if $(\mathbf{A}\mathbf{Q}, s)$, where $\mathbf{Q} \in O(q)$, is another representative of $[\mathbf{A}, s]$, let $(\mathbf{V}', c')$ be the horizontal lift of v to $(\mathbf{A}\mathbf{Q}, s)$. With the function $R_{\mathbf{Q}} : \Theta \rightarrow \Theta$ defined by $R_{\mathbf{Q}}((\mathbf{A}, s)) = (\mathbf{A}\mathbf{Q}, s)$, we have

$$v = d\pi_{(\mathbf{A},s)}(\mathbf{V}, c) = d(\pi \circ R_{\mathbf{Q}})_{(\mathbf{A},s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q},s)}(dR_{\mathbf{Q}})_{(\mathbf{A},s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q},s)}(\mathbf{V}\mathbf{Q}, c).$$

Thus, $\mathbf{V}' = \mathbf{V}\mathbf{Q}$ and $c' = c$. Substitute this into the definition of γ will yield the exact same curve.

The fact that γ is a geodesic follows from a well-known property of geodesics in case of Riemannian submersion (see for example [O'Neill(1967)] Section 2 Corollary 2 or [Valencia(2021)] Theorem 3.6). Any geodesic in Θ with initial velocity being horizontal will always have horizontal velocities and will remain a geodesic in M after applying a Riemannian submersion. $\square$

To compute the logarithmic map, we need first preparatory lemmas that work out an orthogonal matrix based on given points in the space.

Lemma 3. If $\mathbf{L}$ is an invertible $q \times q$ matrix and $\mathbf{Q} \in O(q)$ are such that $\mathbf{L}\mathbf{Q}$ is symmetric, then there exists matrices $\mathbf{R} \in O(q)$, $\mathbf{P} \in O(q)$, a $q \times q$ diagonal matrix $\mathbf{D}$ whose diagonal entries are 1 or -1 , and a $q \times q$ diagonal matrix $\mathbf{S}$ containing the singular values of $\mathbf{L}$ in increasing order such that $\mathbf{L} = \mathbf{R}\mathbf{S}\mathbf{P}^T$ and $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$.

Proof. Let us start by applying the singular value decomposition of $\mathbf{L}$ to get matrices $\mathbf{R}' \in O(q)$, $\mathbf{P}' \in O(q)$ and $\mathbf{S}$ such that $\mathbf{L} = \mathbf{R}'\mathbf{S}(\mathbf{P}')^T$. Substituting the given condition that $\mathbf{L}\mathbf{Q}$ is symmetric will yield that $\mathbf{S}\mathbf{J} = \mathbf{J}^T\mathbf{S}$, where $\mathbf{J} := (\mathbf{P}')^T\mathbf{Q}\mathbf{R}'$.

Clearly $\mathbf{J} \in O(q)$. The equation

$$\mathbf{S}^2 = \mathbf{S}\mathbf{J}\mathbf{J}^T\mathbf{S} = (\mathbf{S}\mathbf{J})(\mathbf{S}\mathbf{J})^T = (\mathbf{J}^T\mathbf{S})(\mathbf{J}^T\mathbf{S})^T = \mathbf{J}^T\mathbf{S}^2\mathbf{J}$$

shows that

$$\mathbf{J}\mathbf{S}^2 = \mathbf{S}^2\mathbf{J}.$$

Looking at each entry of both size, we end up with

$$\mathbf{J}_{ij}(\mathbf{S}_{jj}^2 - \mathbf{S}_{ii}^2) = 0$$

for $i, j = 1, \dots, q$. Given that $\mathbf{L}$ is invertible, its singular values are all strictly positive. Therefore, in fact we have

$$\mathbf{J}_{ij}(\mathbf{S}_{jj} - \mathbf{S}_{ii}) = 0 \quad (1.8)$$

for all i and all j so that $\mathbf{J}\mathbf{S} = \mathbf{S}\mathbf{J} = \mathbf{J}^T\mathbf{S}$. Invert $\mathbf{S}$ yields that $\mathbf{J}$ is actually symmetric. Being a symmetric orthogonal matrix, $\mathbf{J}^2 = \mathbf{I}_q$ and the eigenvalues of $\mathbf{J}$ must be 1 or -1 . Another look at (1.8) shows that $\mathbf{J}$ is a block-diagonal matrix, diagonal if and only if all diagonal entries of $\mathbf{S}$ are distinct. Furthermore, each block of $\mathbf{J}$ must still be orthogonal and symmetric.

We now diagonalize $\mathbf{J}$ based on the above information. Suppose that $\mathbf{S} = \bigoplus_{i=1}^k s_i \mathbf{I}_{m_i}$, where $s_1 > \dots > s_k$ are the unique diagonal entries of $\mathbf{S}$ and each repeats for m_i times. Thus, $\mathbf{J}$ has k blocks. Write the i th block by $\mathbf{J}_i$ so that $\mathbf{J} = \bigoplus_{i=1}^k \mathbf{J}_i$. The spectral theorem then allows us to orthogonally diagonalize $\mathbf{J}_i = \mathbf{K}_i \mathbf{D}_i \mathbf{K}_i^T$, where $\mathbf{K}_i \in O(m_i)$ and $\mathbf{D}_i$ is an $m_i \times m_i$ diagonal matrix with 1 or -1 diagonal entry. Put $\mathbf{K} = \bigoplus_{i=1}^k \mathbf{K}_i$ and $\mathbf{D} = \bigoplus_{i=1}^k \mathbf{D}_i$. Then $\mathbf{K} \in O(q)$, $\mathbf{D}$ is a $q \times q$ diagonal matrix with 1 or -1 diagonal entry, and $\mathbf{J} = \mathbf{K} \mathbf{D} \mathbf{K}^T$. This involved construction leads us to the outcome that $\mathbf{K}$ in fact commutes with $\mathbf{S}$:

$$\mathbf{K}\mathbf{S} = \bigoplus_{i=1}^k s_i \mathbf{K}_i = \mathbf{S}\mathbf{K}.$$

Put $\mathbf{P} = \mathbf{P}'\mathbf{K}$, $\mathbf{R} = \mathbf{R}'\mathbf{K}$. We have

$$\mathbf{Q} = \mathbf{P}'\mathbf{J}(\mathbf{R}')^T = \mathbf{P}'\mathbf{K}\mathbf{D}\mathbf{K}^T(\mathbf{R}')^T = \mathbf{P}\mathbf{D}\mathbf{R}^T$$

and

$$\mathbf{R}\mathbf{S}\mathbf{P}^T = \mathbf{R}'\mathbf{K}\mathbf{S}\mathbf{K}^T(\mathbf{P}')^T = \mathbf{R}'\mathbf{S}\mathbf{K}\mathbf{K}^T(\mathbf{P}')^T = \mathbf{R}'\mathbf{S}(\mathbf{P}')^T = \mathbf{L}.$$

This completes the proof. □

Lemma 4. If $\mathbf{L}$ is an invertible $q \times q$ matrix with singular value decomposition $\mathbf{L} = \mathbf{R}\mathbf{S}\mathbf{P}^T$, then

$$\mathbf{P}\mathbf{R}^T = (\mathbf{L}^T\mathbf{L})^{-\frac{1}{2}}\mathbf{L}^T.$$

This means, although neither $\mathbf{P}$ nor $\mathbf{R}$ is uniquely determined, $\mathbf{P}\mathbf{R}^T$ is uniquely determined.

Proof. Given that

$$\mathbf{L}^T\mathbf{L} = \mathbf{P}\mathbf{S}^2\mathbf{P}^T,$$

we have

$$(\mathbf{L}^T\mathbf{L})^{\frac{1}{2}} = \mathbf{P}\mathbf{S}\mathbf{P}^T.$$

Therefore:

$$\mathbf{L} = \mathbf{R}\mathbf{P}^T\mathbf{P}\mathbf{S}\mathbf{P}^T = \mathbf{R}\mathbf{P}^T(\mathbf{L}^T\mathbf{L})^{\frac{1}{2}}.$$

The result follows from inverting $(\mathbf{L}^T\mathbf{L})^{\frac{1}{2}}$ and take transpose. □

Proposition 5. For every $[\mathbf{A}, s] \in M$ there exists an open neighborhood $U \subset M$ containing $[\mathbf{A}, s]$ such that

1. For every $[\mathbf{B}, t] \in U$, $\mathbf{A}^T\mathbf{B}$ is invertible.

2. The logarithmic map $\text{Exp}_{[\mathbf{A},s]}^{-1} : U \rightarrow T_{[\mathbf{A},s]}M$ is well-defined and is smooth in U . In fact:

$$\text{Exp}_{[\mathbf{A},s]}^{-1}([\mathbf{B}, t]) = d\pi_{(\mathbf{A},s)}((\mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B}) - \mathbf{A}, t - s)),$$

where

$$\mathbf{Q}(\mathbf{A}, \mathbf{B}) = (\mathbf{B}^T \mathbf{A} \mathbf{A}^T \mathbf{B})^{-\frac{1}{2}} \mathbf{B}^T \mathbf{A} \in O(q).$$

Proof. First of all, if $\mathbf{R} \in O(q)$ and $\mathbf{P} \in O(q)$, then $\mathbf{Q}(\mathbf{A}\mathbf{P}, \mathbf{B}\mathbf{R}) = \mathbf{R}^T \mathbf{Q}(\mathbf{A}, \mathbf{B}) \mathbf{P}$ and the same computation as in the proof of Proposition 2 shows that the formula of logarithmic map is indeed well-defined.

Given that the determinant is a continuous function and $\det(\mathbf{A}^T \mathbf{A}) > 0$, there exists a neighborhood $U_\Theta \subset \Theta$ containing $(\mathbf{A}, s)$ such that $\det(\mathbf{A}^T \mathbf{B}) > 0$ whenever $(\mathbf{B}, t) \in U_\Theta$. Thus, for any $[\mathbf{B}, t] \in U_1 := \pi(U_\Theta) \subset M$, $\mathbf{A}^T \mathbf{B}$ is invertible.

Next, it is a well-known fact that $\text{Exp}_{[\mathbf{A},s]}$ is a local diffeomorphism and thus there exists an open neighborhood $U_2 \subset M$ containing $[\mathbf{A}, s]$ such that the logarithmic map $\text{Exp}_{[\mathbf{A},s]}^{-1}$ is well-defined and is smooth in U_2 .

We now compute $\text{Exp}_{[\mathbf{A},s]}^{-1}$ for $[\mathbf{B}, t] \in U := U_1 \cap U_2$. Suppose that $v \in T_{[\mathbf{A},s]}M$ satisfies $v = \text{Exp}_{[\mathbf{A},s]}^{-1}([\mathbf{B}, t])$. By Proposition 2, $[\mathbf{B}, t] = [\mathbf{A} + \mathbf{V}, s + c]$, where $(\mathbf{V}, c)$ is the horizontal lift of v to $(\mathbf{A}, s)$. It follows that $c = t - s$ and that there exists a $\mathbf{Q} \in O(q)$ such that $\mathbf{V} = \mathbf{B}\mathbf{Q} - \mathbf{A}$. Or equivalently:

$$\text{Exp}_{[\mathbf{A},s]}^{-1}([\mathbf{B}, t]) = d\pi_{(\mathbf{A},s)}((\mathbf{B}\mathbf{Q} - \mathbf{A}, t - s)). \quad (1.9)$$

Given that $(\mathbf{V}, c)$ is a horizontal tangent vector, Proposition 1 shows that

$$\mathbf{A}^T(\mathbf{B}\mathbf{Q} - \mathbf{A}) = (\mathbf{B}\mathbf{Q} - \mathbf{A})^T \mathbf{A}$$

and consequently $\mathbf{A}^T \mathbf{B}\mathbf{Q}$ is symmetric. Lemma 3 with $\mathbf{L} = \mathbf{A}^T \mathbf{B}$ shows the existence of $\mathbf{R}, \mathbf{P} \in O(q)$, a diagonal matrix $\mathbf{S}$ with the singular values of $\mathbf{A}^T \mathbf{B}$ in increasing order, and a $q \times q$ diagonal matrix $\mathbf{D}$ with diagonal entries 1 or -1 such that $\mathbf{A}^T \mathbf{B} = \mathbf{R}\mathbf{S}\mathbf{P}^T$ and $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$. (1.9) now becomes

$$\text{Exp}_{[\mathbf{A},s]}^{-1}([\mathbf{B}, t]) = d\pi_{(\mathbf{A},s)}((\mathbf{B}\mathbf{P}\mathbf{D}\mathbf{R}^T - \mathbf{A}, t - s)). \quad (1.10)$$

Now let $\mathbf{B} \rightarrow \mathbf{A}$ and $t \rightarrow s$. The left hand side of (1.10) now becomes $\text{Exp}_{[\mathbf{A},s]}^{-1}([\mathbf{A}, s]) = \mathbf{0} \in T_{[\mathbf{A},s]}M$. The singular value decomposition converges to the orthogonal diagonalization because $\mathbf{A}^T \mathbf{A}$ is symmetric, meaning that $\mathbf{P} = \mathbf{R}$. Thus

$$\mathbf{A}\mathbf{P}\mathbf{D}\mathbf{P}^T - \mathbf{A} = \mathbf{0}.$$

Therefore, $\mathbf{P}\mathbf{D}\mathbf{P}^T = \mathbf{I}_q$ and $\mathbf{D} = \mathbf{I}_q$. Lemma 4 then shows that

$$\mathbf{Q} = \mathbf{P}\mathbf{R}^T = (\mathbf{B}^T \mathbf{A} \mathbf{A}^T \mathbf{B})^{-\frac{1}{2}} \mathbf{B}^T \mathbf{A}.$$

This completes the proof. □

Remark 6. Unfortunately, the logarithmic map is not globally defined in the standard way even over all $[\mathbf{B}, t]$ such that $\mathbf{A}^T \mathbf{B}$ is invertible. For a counterexample, take $p = 2, q = 1, \mathbf{A} = [1, 0]^T, \mathbf{B} = [1, 1]^T$ and $s = t = 1$. In this case, $O(1) = \{1, -1\}$. Then, although $\mathbf{A}^T \mathbf{B} = 1$ is invertible, we still have

$$\text{Exp}_{[\mathbf{A},s]}(d\pi_{(\mathbf{A},s)}((\mathbf{B} - \mathbf{A}, 0))) = [\mathbf{B}, 1] = [-\mathbf{B}, 1] = \text{Exp}_{[\mathbf{A},s]}(d\pi_{(\mathbf{A},s)}((-\mathbf{B} - \mathbf{A}, 0))).$$

The reason is that, even when we assume $\mathbf{A}^T \mathbf{B}$ is invertible, deducing $\mathbf{D} = \mathbf{I}_q$ relies on letting $[\mathbf{B}, t] \rightarrow [\mathbf{A}, s]$. If $[\mathbf{B}, t]$ is far away from $[\mathbf{A}, s]$, then $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$ with some $\mathbf{D} \neq \mathbf{I}_q$ can still produce a valid horizontal vector whose image under $\text{Exp}_{[\mathbf{A},s]}$ is $[\mathbf{B}, t]$, so that $\text{Exp}_{[\mathbf{A},s]}$ fails to be injective and admits no globally defined inverse.

Nevertheless, the choice $D = I_q$ has an interesting property of selecting the geodesic of minimal speed and length when $[B, t]$ is any point such that $A^T B$ is invertible. To see this, consider a general candidate $Q' = PDR^T$ and write $v' = d\pi_{(A,s)}((BQ' - A, t - s))$, so that $(BQ' - A, t - s)$ is the horizontal lift of v' at (A, s) and $\text{Exp}_{[A,s]}(v') = [B, t]$. Let us compute the norm of v' :

$$\begin{aligned} g_{[A,s]}(v', v') &= g_{(A,s)}^\Theta((BQ' - A, t - s), (BQ' - A, t - s)) \\ &= \text{tr}((BQ' - A)(BQ' - A)^T) + (t - s)^2 = \text{tr}(BB^T) + \text{tr}(AA^T) - 2\text{tr}(SD) + (t - s)^2, \end{aligned}$$

where the last equality substituted $A^T BQ' = RSP^T PDR^T = RSDR^T$ and used the cyclic property of the trace. The fact that the diagonal entries of S are strictly positive show that $g_{[A,s]}(v', v')$ is minimized precisely when $D = I_q$. Given that the geodesic starting at $[A, s]$ with initial velocity v' reaches $[B, t]$ at time 1 with constant speed $\sqrt{g_{[A,s]}(v', v')}$, its length is $\sqrt{g_{[A,s]}(v', v')}$ so that $D = I_q$ gives the slowest and hence shortest geodesic. To illustrate, let us go back to the counterexample in the previous paragraph. Put $v_\pm := d\pi_{(A,s)}((\pm B - A, 0))$. Direct computation shows $g_{[A,s]}(v_+, v_+) = 1$ and $g_{[A,s]}(v_-, v_-) = 5$ so that the option corresponding to $D = I_1 = 1$ gives the slowest speed. Therefore, it is actually possible to define $\text{Exp}_{[A,s]}^{-1}$ over all $[B, t]$ such that $A^T B$ is invertible by manually asking it to output the minimal speed candidate and this definition will yield the same $Q(A, B)$ formula.

However, the situation totally breaks down in case $A^T B$ is not invertible. Consider again $p = 2, q = 1$, $A = [1, 0]^T$, $s = t = 1$, but this time $B = [0, 1]^T$. Then $g_{[A,s]}(v_+, v_+) = g_{[A,s]}(v_-, v_-) = 2$ and there are two distinct geodesics of the same speed so $\text{Exp}_{[A,s]}^{-1}([B, t])$ cannot be well-defined.

(Picture: there is a hole at $[0, 0]^T$. Starting $A = [1, 0]^T$, as $B = [0, 1]^T$ is glued to $[0, -1]^2$, v_+ points from $[1, 0]^T$ to $[0, 1]^T$, v_- points from $[1, 0]^T$ to $[0, -1]^T$)

Given that our work should not require a global definition, we stick to the standard definition of logarithmic map only defined locally around $[A, s]$.

1.3 Properties of the Probabilistic Principal Component Analysis Model

We now investigate some properties of the model when the parameter space is now M .

References

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